SMART GLOVE FOR SIGN LANGUAGE TRANSLATION USING IoT

PROJECT REPORT- PHASE 1

Submitted to

Visvesvaraya Technological University

BELAGAVI - 590 018

by

RACHANA JALISATGI	**4SU22EC074**
RASHMI SHETTY	**4SU22EC082**

Under the guidance of

Mrs. Shaileshwari S

Assistant Professor

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering

Department of Electronics & Communication Engineering

SDM INSTITUTE OF TECHNOLOGY

UJIRE – 574240

2024-2025

SDM INSTITUTE OF TECHNOLOGY

(Affiliated to Visvesvaraya Technological University, Belagavi)

UJIRE – 574 240

Department of Electronics & Communication Engineering

CERTIFICATE

Certified that the Project Work titled **'Smart Glove for Sign Language Translation Using IoT'** is carried out by **Ms. RACHANA JALISATGI (4SU22EC074), Ms. RASHMI SHETTY (4SU22EC082)** are bona-fide students of SDM Institute of Technology, Ujire, in partial fulfilment for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during the year 2024-2025. It is certified that all the corrections/ suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The phase 1 report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Mrs. Shaileshwari S	**Dr. Madhusudhana K**	**Dr. Ashok Kumar T**
Asst. Prof. & Guide	HOD	Principal

Table of Contents

List of Figures

List of Tables

List of Acronyms and Abbreviations

CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
ASR	Automatic speech recognition
ISL	Indian Sign Language
ASL	American Sign Language
IMU	Inertial measurement unit
IoT	Internet of things

Chapter 1

INTRODUCTION

In society, most people are not familiar with sign language, which is commonly used by individuals who are deaf or mute, and many are unwilling to learn it. This creates significant challenges for the deaf and mute community when trying to communicate with vocal individuals. To address this issue, we propose a project that will greatly enhance and simplify communication between non-vocal and vocal people.

Signing can be broken down into three major components: stationary gestures, gestures involving movement and facial expressions. However, a majority of the signs can be identified by the handbased gestures. Facial expression and body language are generally more related to the expression of the sentiment underlying the sign. Although, it is worth noticing that the differences in body language from one individual to another, results in a difference in signing from one person to another. Meaning, two people could show variance in signing the same message due to differences in their generic body languages. This is akin to people having different handwritings and for the same word, different people will write differently and there would be variances observed in the final written text [2].

Figure 1.1: Sign gesture for deaf and dumb people

Figure 1.1 shows sign gesture using smart glove which will assist those people who are suffering for any kind of speech defect to communicate through gestures i.e. with the help of single-handed sign language the user will make gestures of alphabets. The glove will record all the gestures made by the user and then it will translate these gestures into visual form as well as in audio form. In this project Flex Sensor Plays the major role, Flex sensors are that change in resistance depending on the amount of bend on the sensor. flex sensors along with accelerometer sensors will track the movement of fingers as well as entire palm [1].

Speech-to-sign language translation involves multiple steps. Initially, the Google Speech API converts spoken language into text, capturing the user's speech input. This text is then processed to identify keywords, phrases, or sentences that correspond to specific sign language gestures. The system compares the extracted text with a database of sign language gestures, which contains mappings between spoken language elements and their corresponding signs. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are commonly used for sign language recognition due to their robustness in handling large datasets and variations in signing styles. These models are trained on extensive datasets to improve accuracy and context understanding [4].

The sign-to-speech converter is a powerful tool designed to support individuals with speech and hearing impairments. It helps convert sign language gestures into spoken words, making communication smoother and more inclusive. This technology bridges the gap between sign language users and those who care taker of them to understand the sign language. By using sensors and Arduino nano, the system captures hand and finger movements. The data is processed and translated into corresponding speech output. This assists users in expressing their thoughts clearly. It promotes independent communication without the constant need for an interpreter. The system can be especially useful in public places, educational institutions, and workplaces. With advancements in IoT, this technology can even be expanded for remote communication. Overall, the sign-to-speech converter plays a crucial role in improving the quality of life for the differently-abled. So, this purpose our goal is to develop smart gloves to translate sign language into speech using IoT.

Chapter 2

LITERATURE REVIEW

2.1 General introduction:

This literature review explores existing research on wearable technology for sign language translation. It highlights the integration of sensors and IoT for real-time gesture recognition and communication. The review also identifies key advancements and challenges in developing accurate, user-friendly systems.

2.2 Literature Survey

2.2.1 Smart Speaking Glove for Deaf and Dumb

The paper titled “Smart Speaking Glove for Deaf and Dumb” was authored by P. Mohan, M. Mohan Raj, M. Kathirvel, P. A. Kasthurirngan, S. Musharaff, and T. Nirmal Kumar. This paper introduces an innovative assistive device a smart glove designed to help individuals with speech and hearing impairments communicate more effectively. The glove uses flex sensors to detect finger movements and gestures, which are then interpreted by an ATmega328 microcontroller. The identified gestures are translated into both visual texts displayed on an LCD screen and audible speech output via a speaker. The system operates in real time and is programmed to recognize a set of predefined hand gestures corresponding to common phrases or words. Additionally, the researchers suggest that the glove could be extended to control home appliances, increasing its utility in daily life. This device aims to bridge the communication gap for people with hearing and speech disabilities, promoting greater inclusivity and independence [1].

2.2.2 Sign Language to Speech Translation

The paper titled “Sign Language to Speech Translation” was authored by Aishwarya Sharma, Dr. Siba Panda, and Prof. Saurav Verma. The paper outlines a system designed to translate “Indian Sign Language (ISL)” into spoken English in real-time. The primary aim of the study is to bridge the communication gap between individuals with hearing or speech impairments and those who do not understand sign language. The system operates by capturing hand gestures through a camera, preprocessing the images using techniques such as segmentation and background subtraction, and recognizing gestures using machine learning algorithms. Once the gesture is identified, it is converted into corresponding text, which is then spoken aloud using text-to-speech technology. The solution proposed in this paper emphasizes low-cost implementation and realtime performance, making it suitable for use in public spaces, educational settings, and personal communication tools. The research contributes significantly to the field of assistive technologies by promoting inclusivity and accessibility for the differently-abled community [2].

2.2.3 Design and Implementation of a sign-to speech/text system for deaf and dumb people

The paper titled “Design and Implementation of a Sign-to-Speech/Text System for Deaf and Dumb People” was authored by Dalal Abdulla, Shahrazad Abdulla, Rameesa Manaf, and Anwar H. Jarndal. This research focuses on developing a “smart glove system” aimed at facilitating communication for individuals who are deaf and mute. The system utilizes “gesture recognition” to translate sign language into “Arabic text and speech”, thereby bridging communication gaps between non-verbal individuals and the hearing community. The glove is interfaced wirelessly with a microcontroller and connected to devices that display text or produce speech. The system has been implemented, programmed, cased, and tested, yielding positive results [3].

2.2.4 Real Time Sign Language Translator Using Machine Learning

The paper titled “Real Time Sign Language Translator Using Machine Learning” was authored by Pradnya Repal, Nikita Pore, Akanksha Pasale, Rutuja Thengal, Swati Ambule, and Prof. N.D. Deval. The study presents a real-time, web-based sign language translation system developed using machine learning techniques to assist individuals with hearing and speech impairments. The system captures hand gestures using a webcam and processes the images through machine learning models built using TensorFlow to recognize and classify the gestures. These gestures are then converted into readable text and further transformed into audible speech using text-to-speech synthesis. The application aims to provide a seamless two-way communication tool, capable of translating both sign language into spoken/written language and vice versa. The project stands out for its versatility, supporting multiple signs and spoken languages, and its practical, user-friendly design that enhances real-time interaction between hearing-impaired users and the wider community. This work is a significant contribution to inclusive technology and digital accessibility [4].

2.2.5 Smart Glove for Sign Language Communications

The paper titled "Smart Glove for Sign Language Communications" and authored by Abhinandan Das and colleagues. This research focuses on developing a wearable smart glove designed to translate sign language gestures into text or speech, thereby facilitating communication for individuals with hearing and speech impairments.

The smart glove integrates flex sensors along each finger to detect finger bending and a MEMS accelerometer to capture hand orientation and movement. These sensors work in tandem to interpret the dynamic gestures of sign language. The collected data is processed using a Recurrent Neural Network (RNN) model, which classifies the gestures into corresponding text or speech outputs. The system was trained on a dataset comprising 27 different classes, enhancing its ability to recognize a diverse set of American Sign Language (ASL) gestures. The study demonstrates the potential of combining sensor technology with machine learning algorithms to create an effective real-time sign language translation tool. Such innovations aim to bridge communication gaps between the hearing and non-hearing communities, promoting inclusivity and accessibility [5].

2.2.6 Machine Learning-Based Gesture Recognition Glove: Design and Implementation

The paper titled "Machine Learning-Based Gesture Recognition Glove: Design and Implementation" by Giovanni Pezzuto, Carmine Zaccagnino, and Giuseppe Valerio Lettieri, presents the development of a low-cost smart glove designed to capture and classify dynamic hand gestures using machine learning techniques. The glove is equipped with five flex sensors, five force sensors, and an inertial measurement unit (IMU), all connected to an ESP32-S2 microcontroller. Data was collected at 100 Hz from multiple participants performing five different gestures. The authors designed and trained a convolutional neural network (CNN) with three convolutional layers to classify these gestures, achieving an accuracy of 90% and an F1-score of 0.9132. The study emphasizes the glove's potential applications in virtual and augmented reality, game control, and assistive technologies for people with disabilities. While the research was limited by the small number of gestures and participants, the results are promising, and the authors suggest future work to improve glove comfort, increase gesture variety, and test the system in realworld scenarios [6].

2.2.7 An IoT-Based Smart Glove Gesture Vocalizer for Deaf and Mute People

The paper titled "An IoT-Based Smart Glove Gesture Vocalizer for Deaf and Mute People", authored by Jyothi R. and Lakshmishri B. L. This study addresses the communication challenges faced by individuals who are deaf and mute by developing a smart glove that translates sign language gestures into audible speech. The proposed system employs flex sensors embedded in a glove to detect hand movements corresponding to specific sign language gestures. These sensor readings are processed by a microcontroller, which interprets the gestures and converts them into vocalized speech using a text-to-speech module. The integration of Internet of Things (IoT) technology allows for real-time communication, enabling users to interact more effectively with others in various environments, including educational, social, and professional settings. This innovation aims to bridge the communication gap between the differently-abled community and the general public, promoting inclusivity and enhancing the quality of life for individuals with hearing and speech impairments [7].

2.2.8 Multi-Purpose Smart Glove for Differently Abled Community People

The paper titled "Multi-Purpose Smart Glove for Differently Abled Community People" authored by Mr. Punit Vora, Mr. Bhushan Parab, Mr. Rohit Bandgar, Mr. Rahul Naik, Dr. Jyoti Dange.This presents
an innovative wearable device designed to assist individuals with disabilities, particularly those who are deaf or hard of hearing. The primary objective of this smart glove is to bridge the communication gap between differently-abled individuals and the general public by translating sign language gestures into both text and speech formats. This functionality is achieved through the integration of flex sensors that detect finger movements, which are then processed by a microcontroller to interpret specific gestures. The interpreted gestures are subsequently converted into audible speech and displayed as text, facilitating effective communication without the need for an interpreter.

The device is designed to be portable and user-friendly, with an accompanying mobile application that assists users in switching between various operational modes. This multifunctional approach not only addresses communication barriers but also supports daily living activities, making it a comprehensive assistive tool for individuals with special needs [8].

2.2.9 Development of a Smart Glove for Sign Language Translation Using Arduino

The paper titled "Development of a Smart Glove for Sign Language Translation Using Arduino," authored by Ahmed J Abougarair, Walaa A Arebi. This presents a project focused on creating an affordable and efficient wearable device to assist communication for the hearing and speech impaired. The smart glove is equipped with flex sensors on the fingers to detect bending movements, an accelerometer on the palm to sense hand orientation, and touch sensors between the fingers to recognize specific contacts. These sensors capture various hand gestures representing sign language, and the data is processed by an Arduino UNO microcontroller. The processed information is then transmitted via Bluetooth to an Android smartphone, where it is converted into readable text and spoken words using a text-to-speech module [9].

2.2.10 Smart Glove for Sign Language Recognition Using Flex and Accelerometer Sensors

The paper titled "Smart Glove for Sign Language Recognition Using Flex and Accelerometer Sensors," authored by Yasir Niaz Khan, Syed Atif Mehdi. This presents the development of a wearable device aimed at assisting communication for individuals with hearing and speech impairments. The smart glove integrates flex sensors attached to the fingers to measure bending and an accelerometer to detect hand orientation and movement. Together, these sensors capture a wide range of sign language gestures, which are processed by a microcontroller to translate the gestures into corresponding letters or words. The output is then displayed as text or converted to speech via a text-to-speech module, enabling real-time communication. Testing showed the system effectively recognizes various gestures with good accuracy, highlighting the smart glove as a practical and non-invasive solution to bridge communication gaps and improve accessibility for the differently-abled community [10].

2.2.11 EchoGlove: Assistive Smart Glove for Gesture-Based Communication and Health Monitoring

The paper titled "EchoGlove: Assistive Smart Glove for Gesture-Based Communication and Health Monitoring" authored by Vijay A, Lochan Pingalay S, Mohanakrishnan. N, Rohan Raghavendra. G, Shalini Agnes Mary A. This presents a comprehensive wearable solution aimed at enhancing communication and safety for individuals with speech and hearing impairments. Developed by researchers from the International Journal for Research Trends and Innovation (IJRTI), EchoGlove integrates multiple functionalities into a single, cost-effective device. The glove utilizes an ESP32 microcontroller to process inputs from various sensors: flex sensors detect specific hand gestures, a pulse sensor monitors heart rate, and an ultrasonic sensor identifies nearby obstacles. Recognized gestures are converted into audible speech via an onboard speaker, facilitating real-time communication. Additionally, the system connects to a mobile application through Bluetooth, displaying the interpreted gestures as text and tracking health metrics, thereby promoting user independence and safety.

The integration of health monitoring features, such as heart rate tracking with alerts for irregularities, adds a critical layer of safety. The obstacle detection capability further assists users in navigating their environment confidently. By combining gesture recognition, speech output, health monitoring, and environmental awareness, EchoGlove stands out as a multifaceted assistive technology. Its affordability and comprehensive feature set make it a promising tool for improving the quality of life for those with communication challenges [11].

2.2.12 Sign Language Translation Using Smart Glove and Mobile Application

The paper titled "Sign Language Translation Using Smart Glove and Mobile Application," authored by Ahmed J Abougarair, Walaa A Arebi.This presents a wearable system designed to facilitate real-time translation of sign language into text and speech. The smart glove incorporates five flex sensors attached to the fingers to detect bending motions and an accelerometer to capture hand orientation and movement. These sensors work in tandem to recognize specific gestures corresponding to the American Sign Language (ASL) alphabet. The captured data is processed by a microcontroller and transmitted via Bluetooth to a mobile application, which then converts the gestures into readable text and synthesized speech. Preliminary experiments demonstrated that the system could translate simple ASL signs with an average response time of approximately 0.6 seconds, highlighting its potential for real-time communication. This integration of hardware and software offers a cost-effective and portable solution to bridge the communication gap between individuals who use sign language and those unfamiliar with it [12].

2.2.13 Wearable Glove for Sign Language Recognition Using Flex and Gyroscope Sensors

The paper titled "Wearable Glove for Sign Language Recognition Using Flex and Gyroscope Sensors," authored by Urav Dalal Aasmi Thadhani, Mahek Upadhye, Shreya Shah, Dr. Meera Narvekar, Dr. Nilesh Patil. This presents the design and implementation of a smart glove aimed at translating sign language gestures into text and speech. The glove integrates five flex sensors, each attached to a finger, to detect bending motions, and an MPU-6050 module that combines an accelerometer and a gyroscope to capture hand orientation and movement. These sensors work in tandem to recognize specific gestures corresponding to the American Sign Language (ASL) alphabet.

The captured data is processed by a microcontroller and transmitted via Bluetooth to a mobile application, which then converts the gestures into readable text and synthesized speech. Preliminary experiments demonstrated that the system could translate simple ASL signs with an average response time of approximately 0.6 seconds, highlighting its potential for real-time communication. This integration of hardware and software offers a cost-effective and portable solution to bridge the communication gap between individuals who use sign language and those unfamiliar with it [13].

2.2.14 Real-Time Sign Language Translation Using Wearable Sensors and Deep Learning Technique

The paper titled "Real-Time Sign Language Translation Using Wearable Sensors and Deep Learning Technique," authored by “This presents a novel system that combines wearable sensor technology with advanced deep learning models to achieve real-time sign language translation. The system features a glove embedded with multiple sensors, including flex sensors to capture finger bending and an accelerometer to detect hand motion and orientation. These sensors generate continuous gesture data, which is fed into a deep learning framework composed of convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to classify and interpret sign language gestures with high accuracy.

The research focuses on translating dynamic hand gestures into textual output in real time, aiming to bridge the communication gap for individuals with hearing or speech impairments. Experimental results demonstrate that the proposed system achieves high gesture recognition accuracy, fast processing speed, and practical usability. The integration of deep learning significantly improves the system’s ability to learn complex gesture patterns and adapt to variations in hand movement. Overall, the study contributes a powerful, responsive, and userfriendly solution for inclusive communication using cutting-edge AI and wearable technology [14].

2.2.15 Deep Learning-Enabled Smart Glove for Real-Time Sign Language Translation

The paper titled "Deep Learning-Enabled Smart Glove for Real-Time Sign Language Translation" by Urav Dalal, Aasmi Thadhani, Mahek Upadhye, Shreya Shah, Meera Narvekar, and Nilesh Patil.This presents the design and development of a wearable smart glove aimed at translating American Sign Language (ASL) gestures into text using deep learning techniques. The glove integrates five flex sensors to capture finger bends and an MPU-6050 sensor to detect hand orientation and motion. These inputs are processed by a Bidirectional Long Short-Term Memory (Bi-LSTM) neural network, which effectively recognizes gesture sequences in real-time. The model is trained on a custom dataset that includes various ASL signs, enabling accurate recognition of both static and dynamic hand movements. The system is designed to aid individuals with speech or hearing impairments by providing a real-time, text-based communication method. Future enhancements include incorporating wireless connectivity, expanding the system to support Indian Sign Language (ISL), and adding a text-to-speech module for voice output. This work demonstrates the promising potential of combining wearable technology with deep learning to bridge communication gaps in inclusive settings [15].

2.2.16 Electronic Speaking Glove for Speechless Patients

The paper titled "Electronic Speaking Glove for Speechless Patients" by Ali, S. Munawwar, and B. Nadeem, focuses on the design and development of a wearable glove system intended to assist speechless patients in communicating. The glove integrates flex sensors to detect finger movements corresponding to specific sign language gestures. These sensor signals are processed through an electronic circuit interfaced with a microcontroller, which translates the gestures into predefined text or speech outputs via a speaker or display unit. The project emphasizes affordability and accessibility, aiming to provide a low-cost assistive device that can enhance the quality of life and communication abilities of individuals with speech impairments. The study highlights the challenges in accurate gesture detection and real-time processing but demonstrates promising results, showing that electronic gloves can be effective tools for sign language translation and communication aid [16].

2.2.17 Captioning for Deaf and Hard of Hearing People by Editing Automatic Speech Recognition in Real Time

The paper titled "Captioning for Deaf and Hard of Hearing People by Editing Automatic Speech Recognition in Real Time" by M. Wald, addresses the challenge of providing accurate and timely captions for deaf and hard-of-hearing individuals. The study explores a system that improves realtime automatic speech recognition (ASR) by incorporating human editing during live transcription, aiming to reduce errors commonly found in automatic captions. This hybrid approach enhances the readability and reliability of captions, enabling better accessibility for users relying on realtime text to follow spoken communication. The paper discusses the technical implementation of the system, its impact on user experience, and the potential to improve communication in various settings such as lectures, meetings, and broadcasts. Overall, this research contributes to advancing assistive technologies that support more inclusive communication for people with hearing impairments [17].

2.2.18 Electronic Hand Glove for Speech Impaired and Paralyzed Patients

The paper "Electronic Hand Glove for Speech Impaired and Paralyzed Patients" by N. P. Bhatti, A. Baqai, B. S. Chowdhry, and M. A. Unar, discusses the design and implementation of a wearable electronic glove aimed at assisting speech-impaired and paralyzed patients in communication. The glove is equipped with flex sensors that detect finger movements corresponding to sign language gestures. These signals are processed by a microcontroller to convert gestures into audible speech or text output, providing an effective communication bridge for individuals unable to speak or move freely. The paper emphasizes the device’s potential to improve the quality of life and social interaction for patients with disabilities by enabling them to express themselves more easily. Additionally, the authors highlight the system's cost-effectiveness and portability, making it a practical solution for real-world applications, especially in resource-limited settings [18].

2.2.19 Design and Development of a Smart Glove for Assistive Communication

The paper titled "Design and Development of a Smart Glove for Assistive Communication" by Christopher Harris and Samantha Carter, presented at the IEEE International Conference on Rehabilitation Robotics (ICORR), focuses on creating a wearable smart glove aimed at improving communication for individuals with speech and motor impairments. The glove is equipped with an array of sensors, including flex and inertial measurement units (IMUs), to accurately capture hand and finger movements corresponding to sign language gestures. Using advanced signal processing and machine learning algorithms, the system translates these gestures into spoken or written language in real time. The study highlights the integration of wearable technology with rehabilitation robotics principles to provide an intuitive and non-invasive communication aid.

Results demonstrate the device's potential for high accuracy and user comfort, supporting its use in daily life for patients requiring assistive communication solutions. This research contributes to the broader effort of developing accessible, effective technologies for enhancing independence among people with disabilities [19].

2.2.20 Integration of Machine Learning Algorithms in Gesture Recognition for Assistive Devices

The paper titled "Integration of Machine Learning Algorithms in Gesture Recognition for Assistive Devices" by Daniel Robinson and Rachel White, explores the application of advanced machine learning techniques to enhance gesture recognition systems used in assistive devices. The study focuses on improving the accuracy and responsiveness of wearable technologies, such as smart gloves, that interpret hand and finger movements for communication and control purposes. By integrating algorithms like convolutional neural networks (CNNs) and support vector machines (SVMs), the authors demonstrate significant improvements in recognizing complex gestures across diverse users and environments. The research also addresses challenges related to real-time processing and variability in human motion, proposing robust models that adapt to individual differences. Overall, the paper contributes valuable insights into leveraging machine learning to make assistive devices more intuitive and effective for users with physical impairments [20].

2.3 Summary:

The literature shows growing interest in smart gloves for sign language translation. Most designs use flex sensors and accelerometers to detect gestures. However, many lack real-time communication and IoT features. Recent tech advancements support more efficient, portable solutions. This project builds on past work to create an IoT-based, real-time translation glove.

Chapter 3

PROBLEM STATEMENT AND OBJECTIVE

3.1 Problem statement:

Lack of widespread sign language understanding creates communication barriers between deaf or hard-of-hearing individuals and the general public, requiring mediation to ensure their rights and messages are conveyed accurately.

3.2 Objective:

- To develop a wearable smart glove capable of detecting hand and finger movements corresponding to sign language gestures.
- To integrate sensor (flex) for accurate real-time gesture recognition.
- To process and translate gestures into readable text or audible speech.

Chapter 4

METHODOLOGY

Figure 4.1 block diagram illustrates a wearable sensor-based system designed using an Arduino Nano as the central processing unit. The system incorporates four flex sensors and an accelerometer to capture motion and gesture data. These sensors are connected to the Arduino Nano, which processes the input signals and transmits the information to external platforms such as a mobile application and IoT-based services for further analysis or action. A power supply module provides the necessary electrical power to the system. This setup is commonly used in applications such as gesture recognition, health monitoring, smart gloves, or assistive communication devices.

4.1 Block diagram

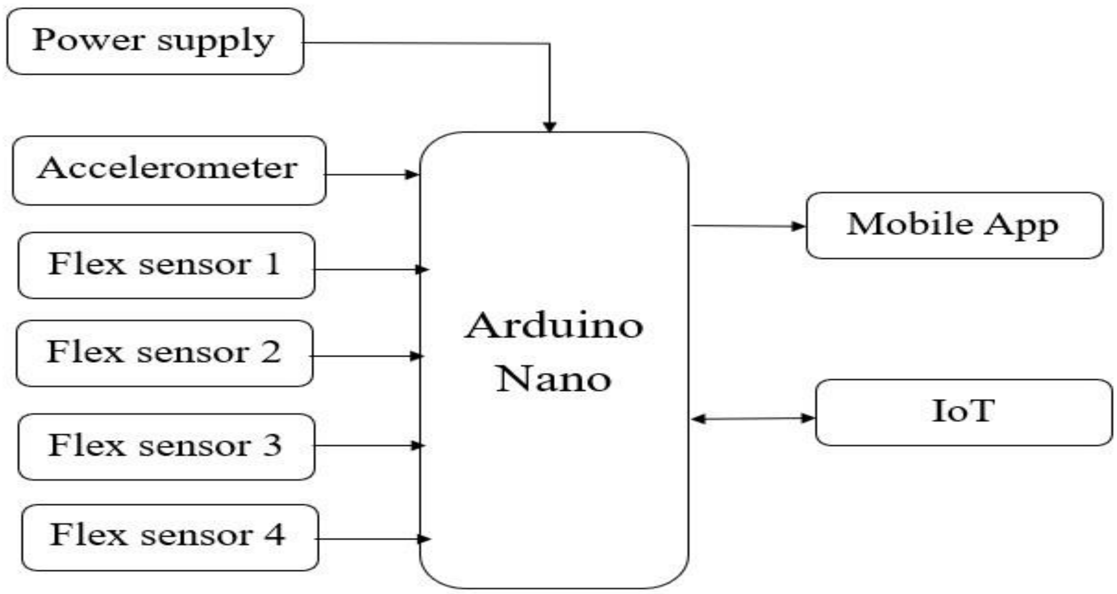


Figure 4.1: Block diagram

- Power Supply: Provides required voltage to all components.
- Accelerometer: Detects motion and orientation of the system.
- Flex Sensor 1-4: Measure bending of fingers or joints.
- Arduino Nano: Processes sensor inputs and controls outputs.
- Speaker: Outputs sound based on Arduino signals.

- IoT Module: Sends or receives data over the internet.

The development of the Smart Glove for Sign Language Translation is divided into several key phases to ensure systematic design, implementation, and testing:

1. System Initialization

- Power is supplied to the entire circuit including the Arduino Nano, sensors, and peripheral devices.
- The Arduino Nano initializes all required modules including the Analog and Digital I/O pins, communication protocols (e.g., Serial, I2C), and connected components.

2. Sensor Data Acquisition

- Flex Sensors (1–4): These sensors are placed on fingers to measure the degree of bending. Each sensor provides analog input corresponding to finger flexion.
- Accelerometer: Captures motion and orientation data of the hand in 3D space. This helps in detecting hand gestures or movement direction.

3. Data Processing (Arduino Nano)

- The Arduino reads analog signals from the flex sensors and the accelerometer.
- The sensor data is analysed to determine specific hand gestures or postures.
- Based on pre-programmed logic, gestures are mapped to specific outputs or actions.

4. Output Generation

- Speaker Output: Upon gesture recognition, the Arduino sends corresponding digital signals to a speaker module to produce voice feedback or sound.
- IoT Communication: The Arduino transmits recognized gesture data to an IoT module (e.g., ESP8266 or ESP32) for cloud storage, remote monitoring, or interaction with other smart devices.

5. IoT Integration

- The IoT module sends real-time gesture data to a remote server or mobile application.
- The system may include feedback from the cloud (e.g., alerts or controls) that can influence Arduino behaviour.

6. Continuous Monitoring

- The system operates in a continuous loop, constantly monitoring sensor inputs, processing data, and responding accordingly.
- It can also log data or trigger alerts based on specific gesture patterns or movements.

4.2 Hardware Components

- **Arduino Nano**
- **Flex sensor – 2'2 inch**
- **Accelerometer**
- **IoT Module**
- **Mobile App**

4.2.1 Arduino Nano

Figure 4.2 represents the Arduino Nano is a compact and versatile microcontroller board based on the ATmega328P, widely used in electronics projects due to its small size and powerful features. It operates at 5V, with 14 digital input/output pins (six of which can be used as PWM outputs), eight analog inputs, and supports standard communication protocols like UART, SPI, and I2C. The Nano includes 32KB of flash memory for storing code, 2KB of SRAM, and 1KB of EEPROM. Its 16 MHz clock speed ensures smooth and responsive performance for a wide range of applications. Designed to be breadboard-friendly, it connects via a Mini USB port for programming and power. Ideal for embedded systems, DIY electronics, sensor monitoring, and robotics, the Arduino Nano offers a cost-effective and reliable platform for both beginners and experienced developers. Additionally, newer variants like the Nano Every, Nano 33 IoT, and Nano 33 BLE expand its capabilities with modern microcontrollers, Wi-Fi, Bluetooth, and onboard sensors, making the Nano family suitable for everything from simple LED projects to complex IoT applications.

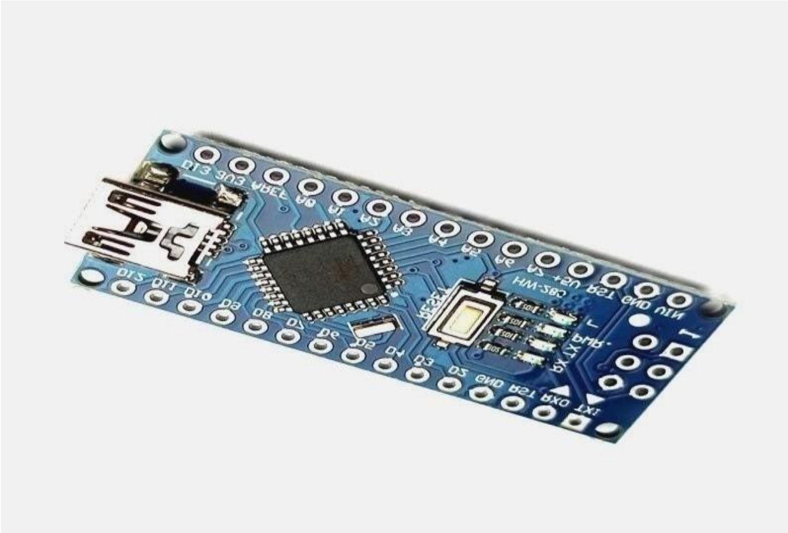


Figure 4.2: Arduino Nano

4.2.2 Flex Sensor

Figure 4.3 represents Flex sensors which are piezoresistive sensors, meaning that the resistance varies depending on the bend's direction and degree of bend. Since each symptom is a variation on the joints, a flex sensor has been implanted on each joint to detect bends. The best results are obtained when flex sensors and a voltage divider design are used. This allows abrupt fluctuations to be handled by a custom filter while suppressing fluctuating values into acceptable ranges. The output of this configuration is sent to the ADC. Since the ADC can tolerate 5 volts, the voltage divider design employs 5 volts input to guarantee coherence between the sensors and the ADC.

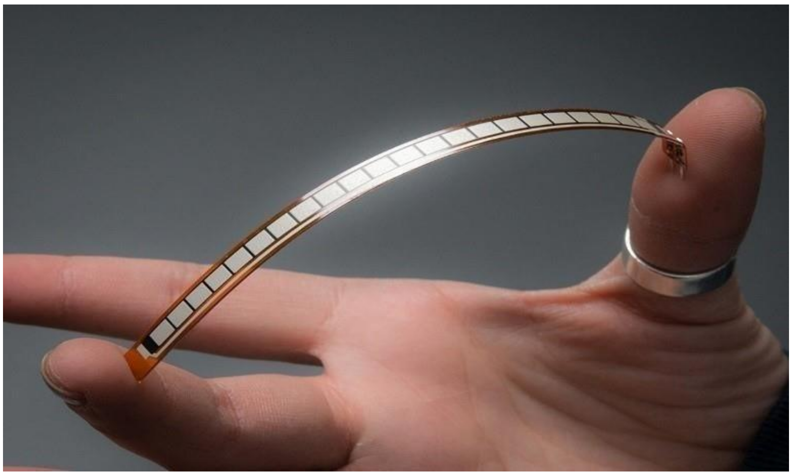


Figure 4.3: Flex Sensor

Features	Specification
Type	Resistive (Passive sensor)
Resistance (Flat)	10K ohm
Resistance (Fully Bent)	30-50K ohm
Length	2.2 inches
Thickness	0.43 mm
Operating voltage	3.3-5 V
Width	0.25 inches
Response Time	40 ms
Angle Range	0-90 degree

Table 1: Features and specification of Flex sensor

4.2.3 Accelerometer

An I2C and SPI interface is available on the low-power, three-axis MEMS accelerometer module, ADXL345. These modules' Adafruit Breakout boards have built-in 3.3-volt voltage regulation and level shifting, which makes interacting with 5-volt microcontrollers like the Arduino straightforward. Figure 4.4 represents the accelerometer.

Figure 4.4: Accelerometer

Features	Specification
Measurement Axes	X, Y, Z (3 Axis)
Output type	Analog Voltage
Voltage supply	3.3-5 V
Output range	0.3-2.7 V per axis
Acceleration range	3g
Current consumption	350 MA
Interface to Arduino	Analog pins (A4, A5, A6, etc)

Table 2: Features and Specification of accelerometer

4.2.4 IoT Module

The IoT module connects the Arduino Nano to the internet. Common modules include ESP8266 or ESP32. It enables the transmission of sensor data to cloud platforms where it can be stored, analysed, or monitored from anywhere in the world. This is particularly useful for remote health monitoring, smart devices, or data logging in research applications. Figure 4.5 represents the IoT Module.

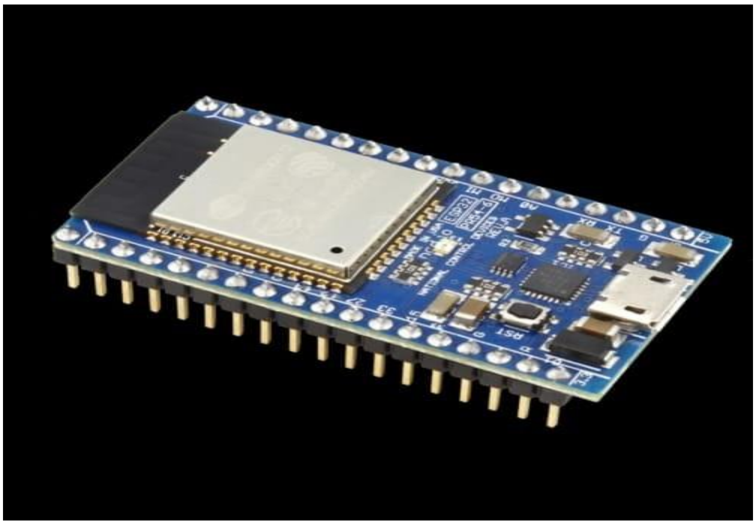
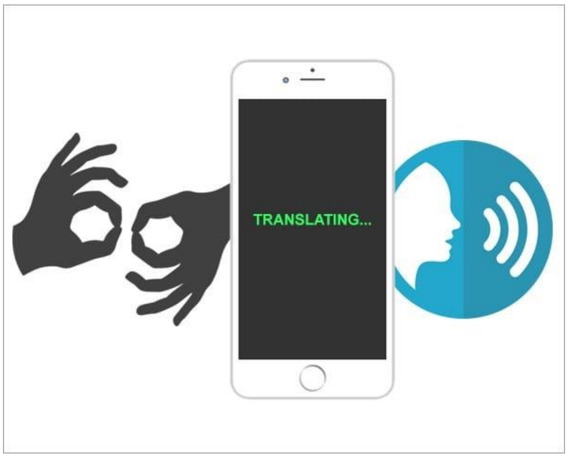


Figure 4.5: IoT Module

4.2.6 Mobile App

The mobile app acts as the user interface for this system. It receives processed data from the Arduino Nano, usually via Bluetooth or Wi-Fi. The app can display real-time sensor values, recognize gestures, and even provide feedback or alerts. It is especially useful for monitoring or controlling the system remotely through a smartphone. Figure 4.6 represents the Mobile App.



.

Figure 4.6: Mobile App

4.3 Software Requirements

- Arduino IDE
- Embedded C

4.3.1 Arduino IDE

The Arduino Integrated Development Environment – or Arduino Software (IDE)- contains a text editor for writing code, a messages area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them. Figure 4.8 represents Arduino IDE.

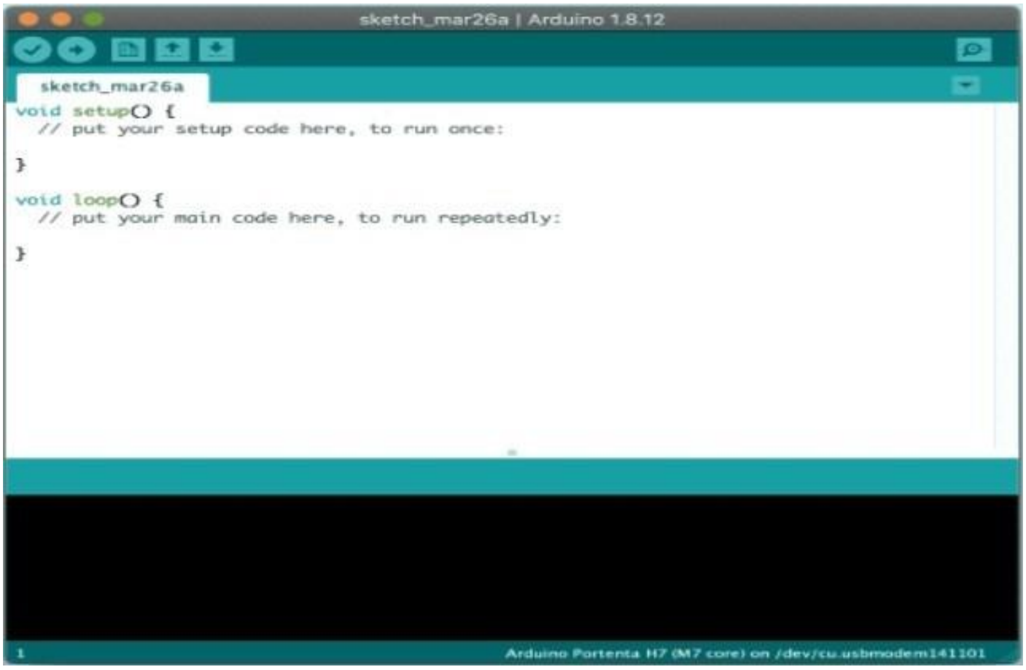


Figure 4.7: Arduino IDE

4.3.2 Embedded C

Embedded C programming plays a crucial role in making the microcontroller run and perform the fancied actions. At present, we usually utilize several electronic devices like mobile phones, writing machines, security systems, refrigerators, digital cameras, etc. The managing of these embedded devices can be done with the help of an embedded C program. Embedded C programming strengthens with functions where every function is a set of statements utilized to complete specific tasks. The embedded C and C languages are equivalent and implemented through significant components like a variable, character set, keywords, data types, declaration of variables, expressions, and statements. All features play a significant role in writing an embedded C program.

PROGRESS TILL DATE

Our project is Smart Gloves for Sign Language Translation using IoT for deaf and dumb people. We have completed the software part and uploaded the code to the Arduino Nano. To ensure the accuracy of our program, we employed the Wokwi app for simulation and testing purposes. The system employs flex sensors and an accelerometer to recognize hand gestures and transmit predefined messages such as “I need help” or “Move right”.

In the software, the Arduino Nano processes signals from the flex sensor to recognize predefined messages such as "I need help”, “Move right” etc. Instructions are programmed to capture and detect hand gestures across multiple axes using the accelerometer. The code integrates these dynamic motion readings with the static gesture data from the flex sensor, allowing the system to accurately interpret and transmit the user's intended commands. This integrated processing logic is implemented and thoroughly tested within the software, with its accuracy validated through simulation.

5.1 Simulation Result

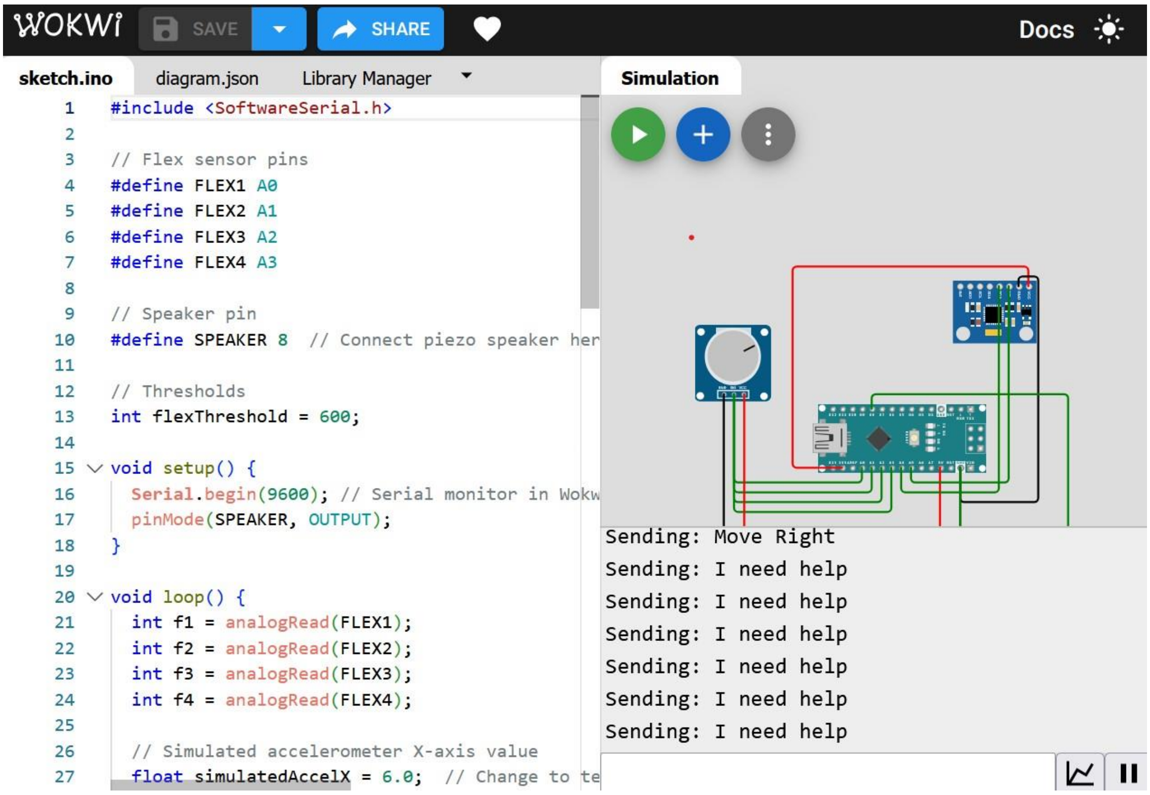


Figure 5.1: Simulation of the Smart Glove for Sign Language Translation

Chapter 6

PLAN FOR COMPLETION OF PENDING WORK

To complete the pending hardware part of our smart glove for sign language translation, begin by collecting all necessary components: four flex sensors, an accelerometer module, an Arduino Nano, an IoT module, a speaker, a suitable power supply and a glove. Attach the flex sensors securely along the fingers of the glove using stitching or glue, and mount the accelerometer on the back of the glove to capture hand orientation. Connect the flex sensors to the analog pins of the Arduino using voltage divider circuits, and wire the accelerometer. The IoT module should be connected via software serial, and the speaker module to one of the digital pins. Ensure all components share a common ground, and the power supply is sufficient and safely connected to the system.

After wiring, upload our completed software code to the Arduino Nano and begin testing. Confirm that all sensors are properly responding flex sensors should vary analog values with finger bending, and the accelerometer should detect motion and orientation changes. Make sure the IoT module can transmit data correctly to the associated app or server, and the speaker produces sound for recognized gestures. Calibrate the sensors for accuracy, then insulate and secure all connections on the glove to make it wearable and durable. Once the glove is tested and functioning smoothly, it will be ready for demonstration or real-world use as a functional sign language translation device.

CONCLUSION

The proposed system effectively demonstrates the integration of flex sensors, an accelerometer, and IoT technology using an Arduino Nano to recognize and respond to hand gestures. By capturing finger movement through flex sensors and hand orientation through an accelerometer, the system can accurately interpret gestures and translate them into meaningful outputs such as audio feedback via a speaker or remote updates through an IoT module. This approach showcases a low-cost, efficient, and portable solution that can be applied in various fields including assistive technology for the speech-impaired, gesture-controlled devices, and smart wearable systems. The successful implementation of this system confirms its potential for real-world applications requiring gesture-based human-machine interaction.

REFERENCES

[1] P. Mohan, M. Mohan Raj, M. Kathirvel P. A. Kasthurirngan, S. Musharaff, T. Nirmal Kumar. "Smart Speaking Glove for Deaf and Dumb".

[2] Aishwarya Sharma, Dr. Siba Panda, Prof. Saurav Verma. "Sign Language to Speech Translation".

[3] D Abdulla, S Manav, & Jarndal A H. "Design and implementation of a sign-to speech/text system for deaf and dumb people".

[4] Ms. Pradnya Repal, Nikita Pore, Akanksha Pasale, Rutuja Thengal, Swati Ambule, and Prof. N.D. Deval. "Real Time Sign Language Translator Using Machine Learning".

[5] Abhinandan Das and colleagues. "Smart Glove for Sign Language Communications".

[6] Giovanni Pezzuto, Carmine Zaccagnino, and Giuseppe Valerio Lettieri. "Machine LearningBased Gesture Recognition Glove: Design and Implementation".

[7] Jyothi R. and Lakshmishri B. L. "An IoT-Based Smart Glove Gesture Vocalizer for Deaf and Mute People".

[8] Mr. Punit Vora, Mr. Bhushan Parab, Mr. Rohit Bandgar, Mr. Rahul Naik, Dr. Jyoti Dange. "MultiPurpose Smart Glove for Differently Abled Community People".

[9] Ahmed J Abougarair, Walaa A Arebi. "Development of a Smart Glove for Sign Language Translation Using Arduino".

[10] Yasir Niaz Khan, Syed Atif Mehdi. "Smart Glove for Sign Language Recognition Using Flex and Accelerometer Sensors".

[11] Vijay A, Lochan Pingalay S, Mohana krishnan N, Rohan Raghavendra G, Shalini Agnes Mary A. "EchoGlove: Assistive Smart Glove for Gesture-Based Communication and Health Monitoring".

[12] Ahmed J Abougarair, Walaa A Arebi. "Sign Language Translation Using Smart Glove and Mobile Application"

[13] Urav Dalal Aasmi Thadhani, Mahek Upadhye, Shreya Shah, Dr. Meera Narvekar, Dr. Nilesh Patil. "Wearable Glove for Sign Language Recognition Using Flex and Gyroscope Sensors ".

[14] Abhishek Wahane, Rishikesh Gadade, Aditya Hundekari, Aditya Khochare, Chudaman Sukte. "Real-Time Sign Language Translation Using Wearable Sensors and Deep Learning Technique"

[15] Urav Dalal, Aasmi Thadhani, Mahek Upadhye, Shreya Shah, Meera Narvekar, and Nilesh Patil. "Deep Learning-Enabled Smart Glove for Real-Time Sign Language Translation"

[16] Ali, S. Munawwar, and B. Nadeem. "Electronic Speaking Glove for Speechless Patients".

[17] M. Wald. "Captioning for Deaf and Hard of Hearing People by Editing Automatic Speech Recognition in Real Time".

[18] N. P. Bhatti, A. Baqai, B. S. Chowdhry, and M. A. Unar. "Electronic Hand Glove for Speech Impaired and Paralyzed Patients".

[19] Christopher Harris and Samantha Carter. "Design and Development of a Smart Glove for Assistive Communication".

[20] Daniel Robinson and Rachel White. "Integration of Machine Learning Algorithms in Gesture Recognition for Assistive Devices".